

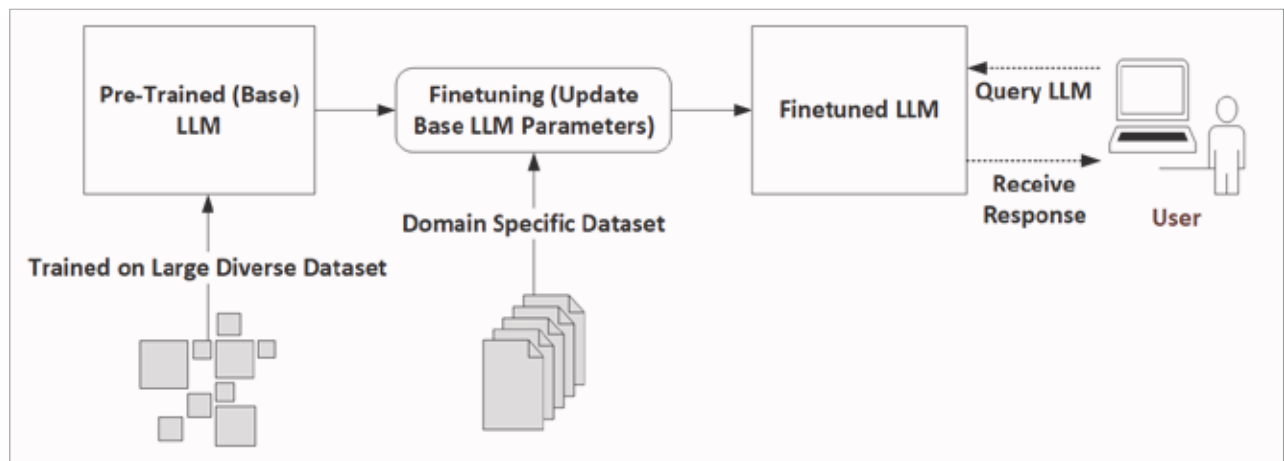


Figure 1: LLM fine-tuning

model's parameters are adjusted so it can learn the domain's terminology, style, and reasoning patterns. This focused training helps the model go beyond generic knowledge and develop a more expert understanding of how information is expressed and interpreted in that field.

Fine-tuning is required over prompt engineering and RAG as:

- Prompt engineering can guide a model but cannot force it to learn domain terminology or internal workflows it has never seen.
- RAG augments the model with retrieved documents, but it doesn't change model behaviour.
- Prompts and retrieved documents cannot guarantee deterministic and compliant behaviour. Fine-tuning allows embedding organisational compliance directly into the model parameters.

Types of fine-tuning

Based on the training objective and kind of data used, fine tuning can be categorised as unsupervised fine-tuning, supervised fine-tuning (SFT) and instruction fine-tuning (IFT).

Unsupervised fine-tuning:

Unsupervised fine-tuning involves training the model further on raw, unlabelled text from a specific domain. It does not teach the model specific

tasks; instead, it strengthens domain understanding, reduces hallucinations, and improves fluency.

Example use cases are:

- Adapting a general LLM to understand medical literature.
- Training on legal contracts to learn clause structure, legal terminology, and formal writing style.

Supervised fine-tuning (SFT): The goal of SFT is to teach the model how to perform a specific task using labelled input-output pairs. Each training example consists of an input prompt and a desired response. The focus is on learning patterns, formats, and decision logic for a particular workflow.

Unlike unsupervised fine-tuning, SFT explicitly guides the model towards correct behaviour and output generation for well-defined workflows.

Example use cases are:

- Training a code-generation model using problem-solution pairs.
- Adapting a model for news classification.

Instruction fine-tuning (IFT):

Instruction fine-tuning (IFT) is a specialised form of supervised fine-tuning in which the input is an explicit natural language instruction, followed by the expected response. The goal of IFT is to train the model to understand and follow human instructions reliably across a wide range of tasks.

IFT improves conversational ability, task compliance, and reasoning behaviour, making it a key technique behind modern chat-based LLMs. Popular open source models such as LLaMA-Instruct and Mistral-Instruct rely heavily on IFT. IFT is typically the final fine-tuning stage before deploying an LLM for interactive applications.

Example use cases are:

- Training a chatbot to respond in a specific tone or writing style.
- Training the model to respond safely and refuse invalid or harmful requests.

Techniques of fine-tuning

Fine-tuning methods can be divided into two broad technical approaches depending on how much of the model is updated and the computational cost involved. These approaches are full fine-tuning and parameter-efficient fine-tuning (PEFT).

Full fine-tuning: Full fine-tuning involves updating all the parameters of the pre-trained LLM during training. The entire LLM (billions of parameters) is unfrozen and learns the new domain end-to-end. It provides the highest level of adaptation and can significantly change model behaviour. It requires large compute resources and substantial training data. However, it can carry risks of overfitting and